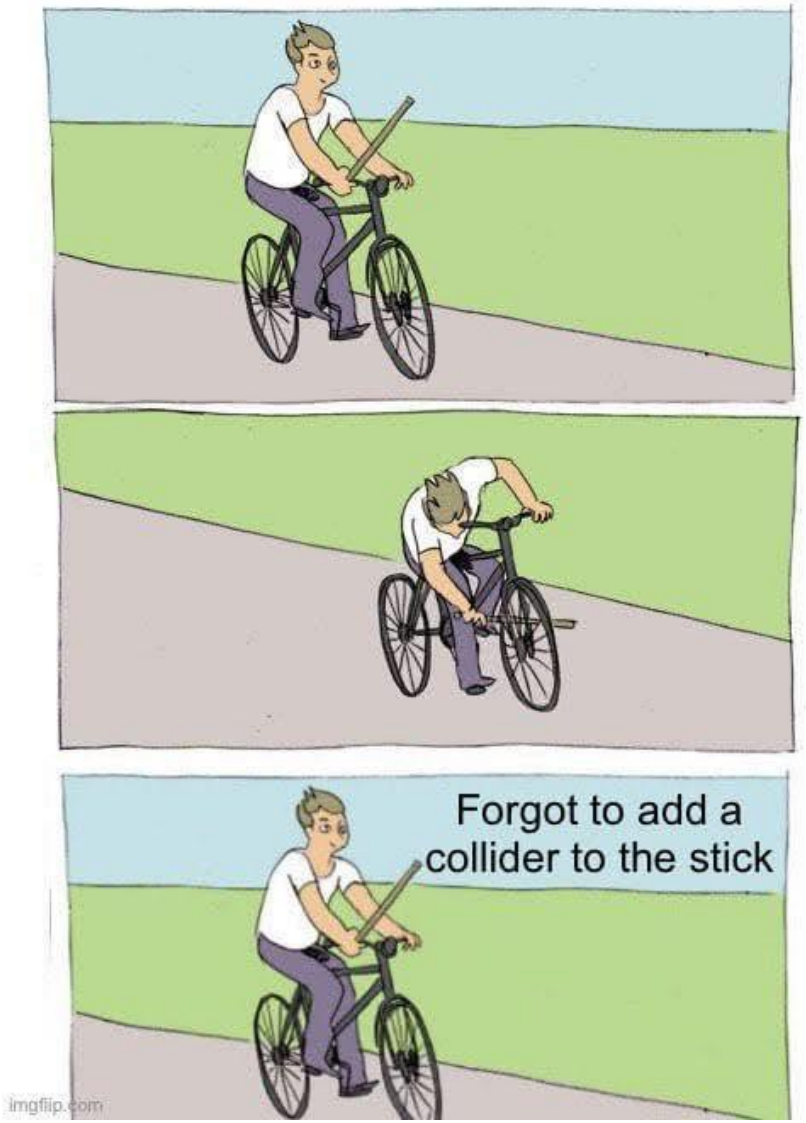
Game Physics Programming: The Art of Not Touching

Spring 2026

School of Computing and Data Science

Wentworth Institute of Technology



Previously

- Circle-Circle Collision and Resolution
- AABB collisions (Reminder in a moment)
- Our math library (including projections and normals)

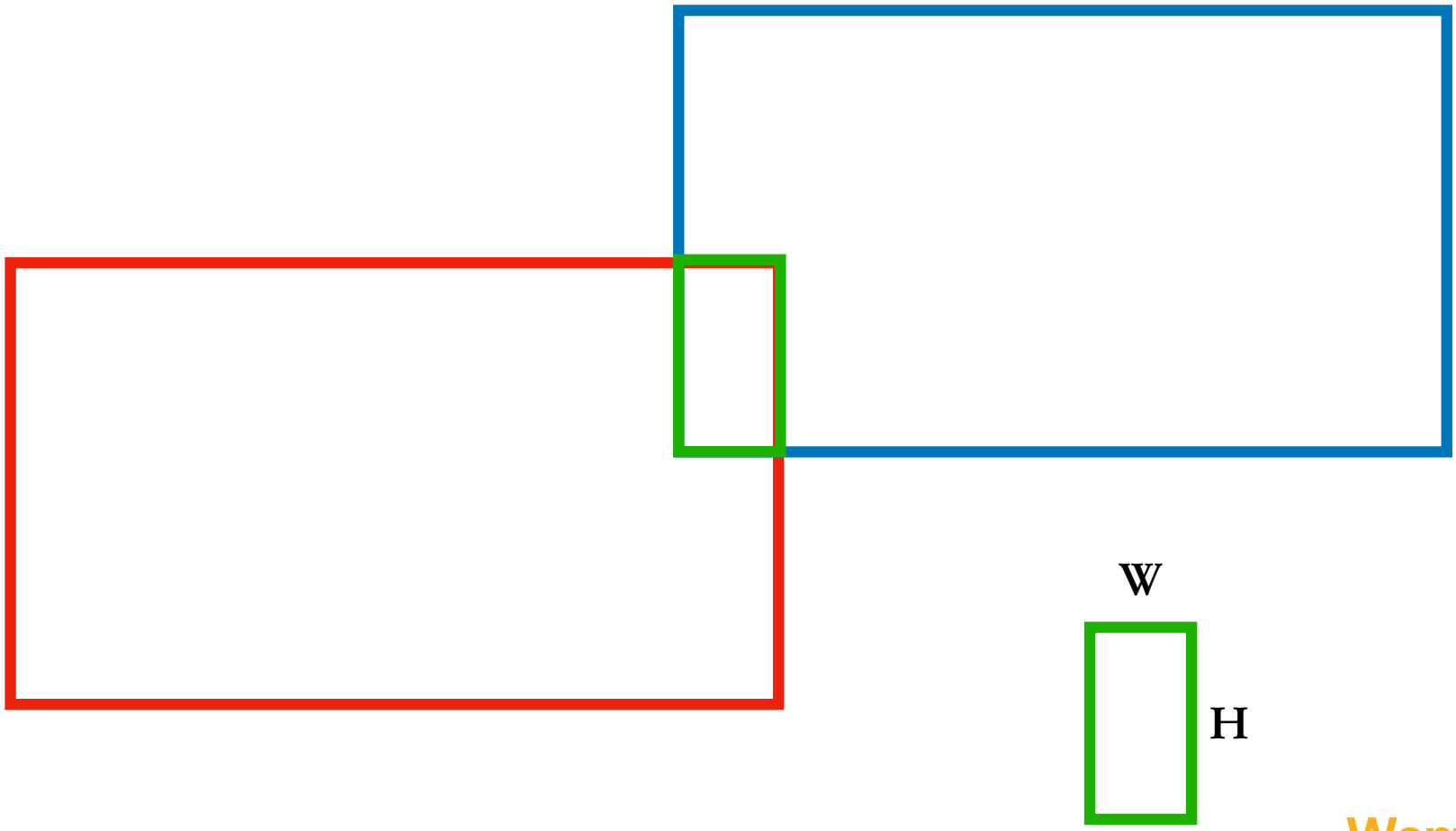
I'll do some live-coding today. I've prepared a GitHub repo if you want to follow along:

https://github.com/mdschuster/GPP-SAT-Toy

It's a toy program that is only meant to illustrate SAT in a simple environment.

The Collision Problem (Simplifed)

- Given two entities that have a current position (only) do they intersect?

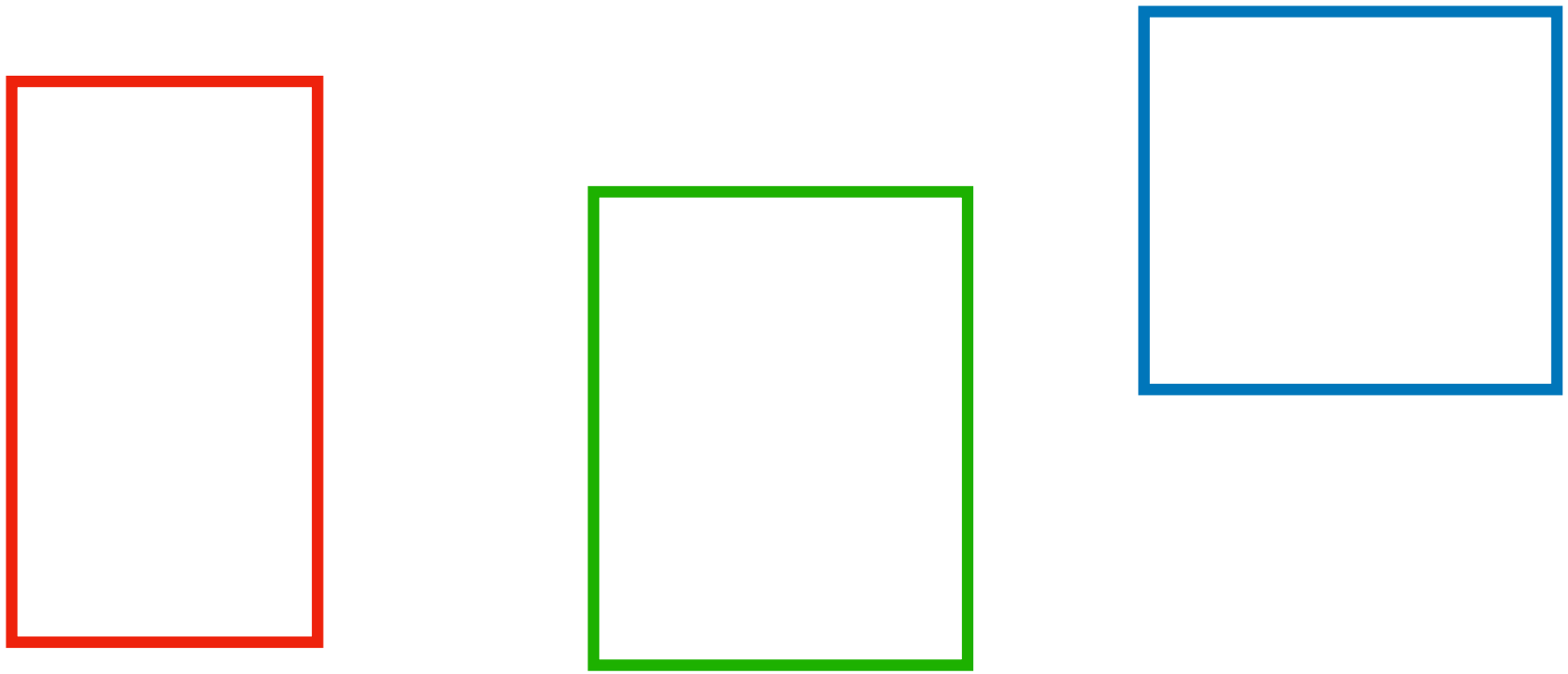


2D Bounding Shapes

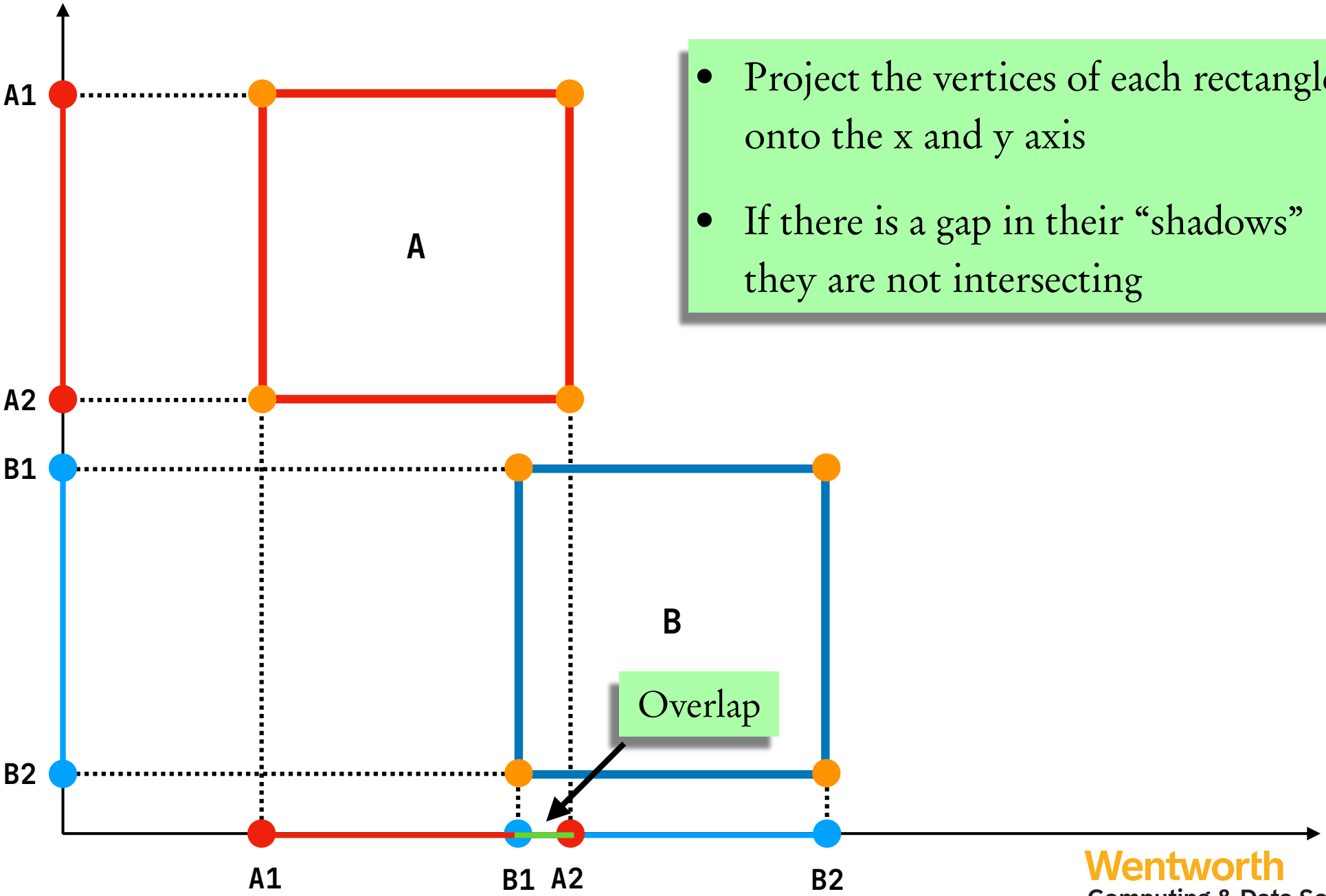
- The simplest 2D bounding shape is the circle.
- We already covered circle intersection
- Other bounding shapes can be constructed with line segments
- The simplest of these is a rectangle
  - Bounding Rectangle/Bounding Box
  - The intersection math is easy, but not trivial
  - Cheap to perform

Reminder: Axis Aligned Bounding Boxes

- By aligning our rectangles with the xy axis, we can do very fast collision checks
- Axis Aligned Bounding Box: **AABB**



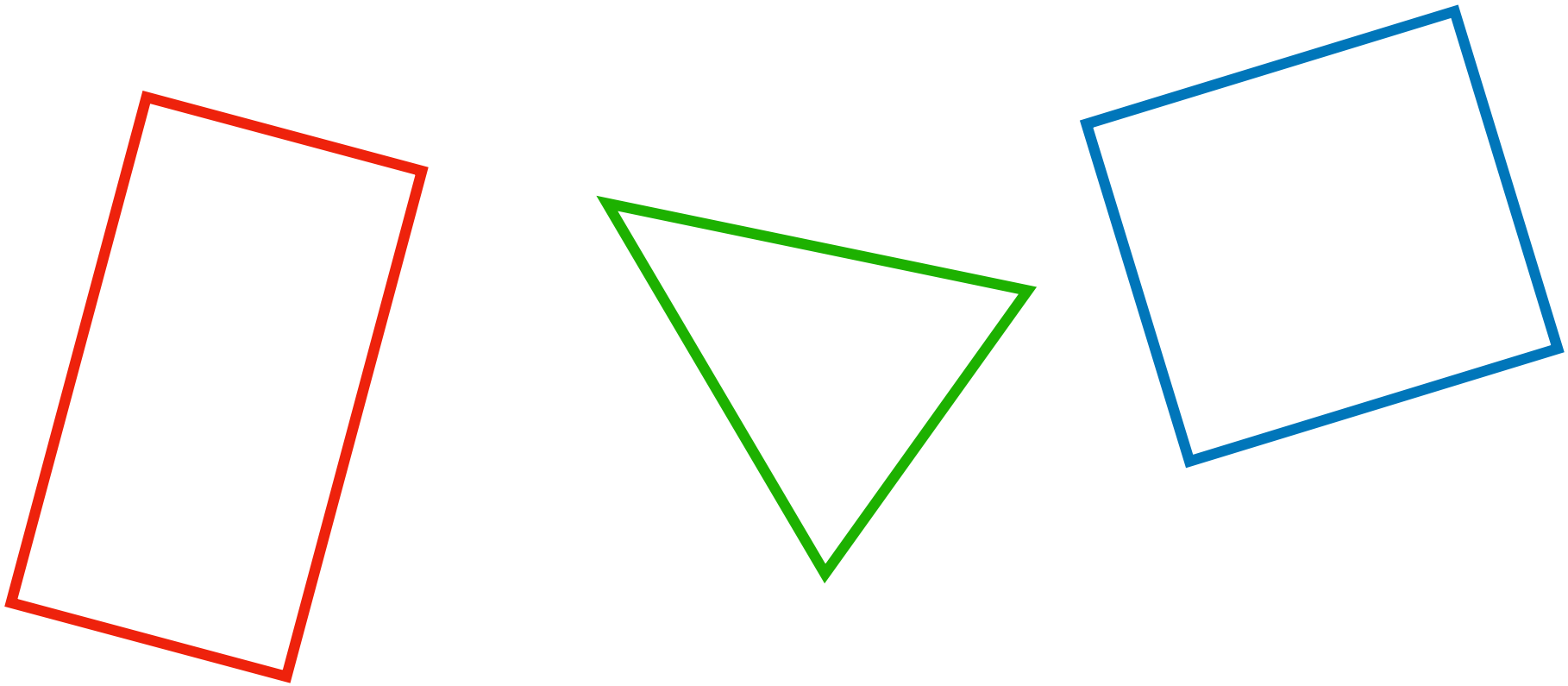
Reminder: AABB



- Project the vertices of each rectangle onto the x and y axis
- If there is a gap in their "shadows" they are not intersecting

What if we want more than AABB?

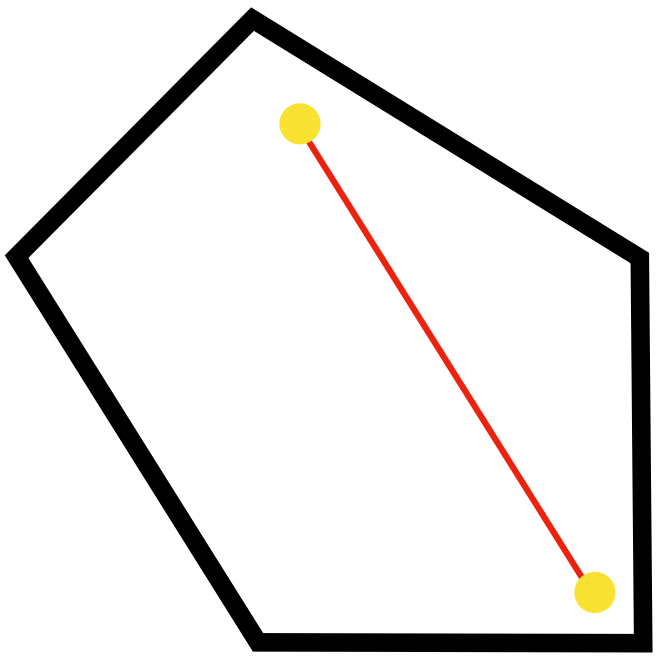
- Rectangles (or any polygon) in 2D can take any orientation:



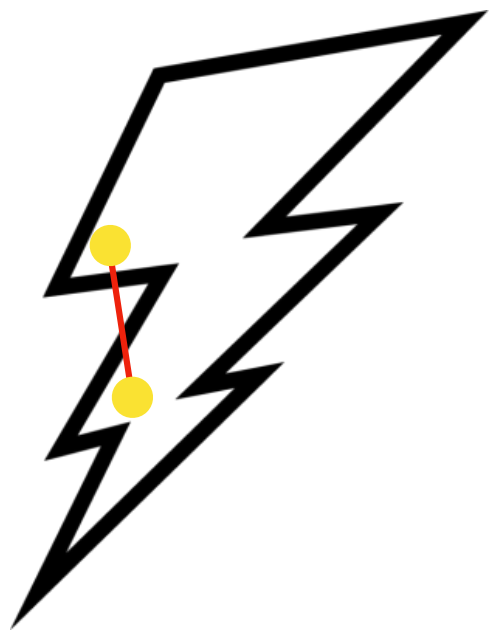
Separating Axis Theorem (SAT)

- Determines if two convex shapes are intersecting
- Can find the minimum intersection vector too

Convex:

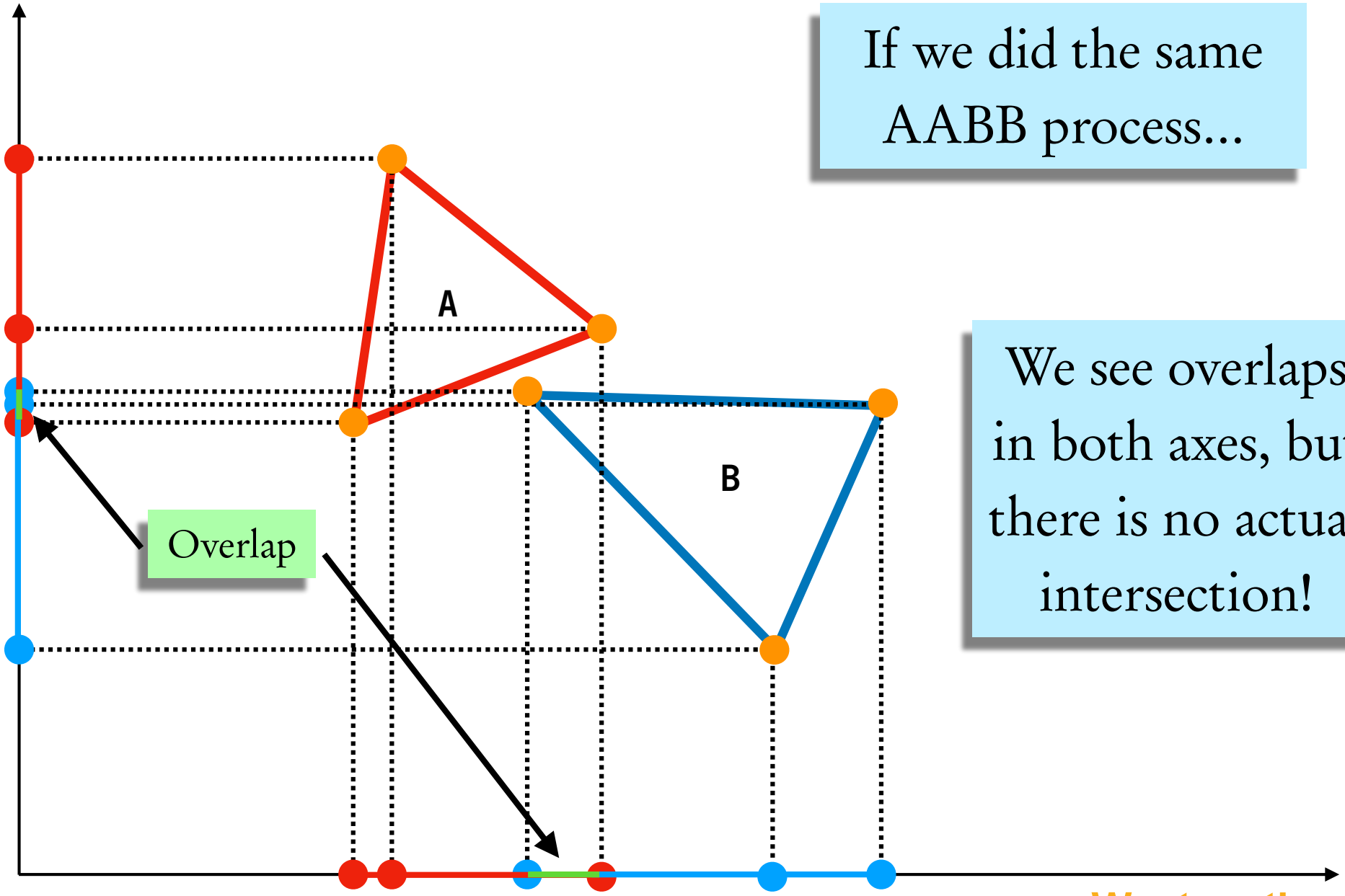


Concave:



A line drawn from any two points in a convex shape will not leave the bounds of the shape

SAT



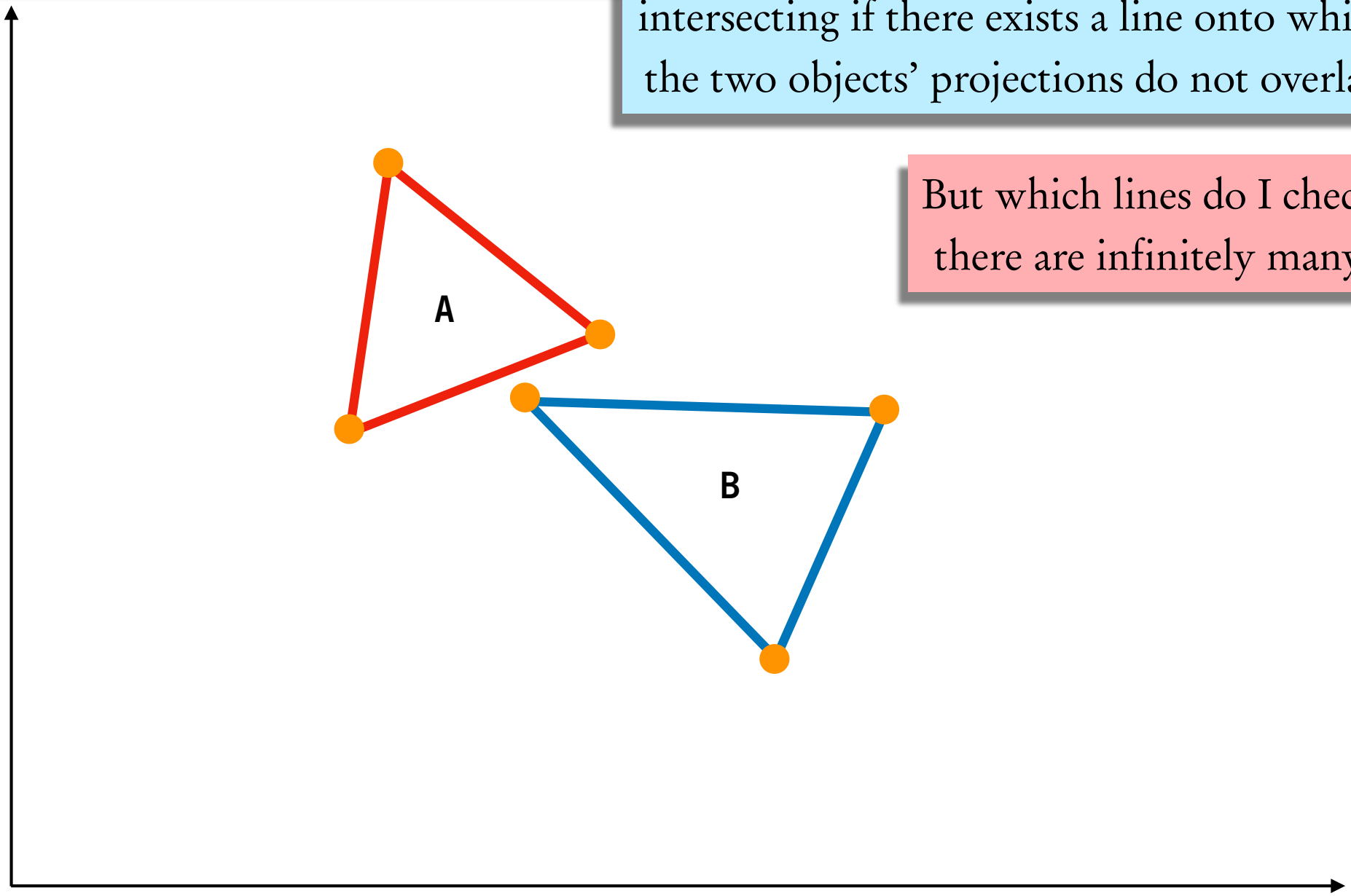
If we did the same AABB process...

We see overlaps in both axes, but there is no actual intersection!

SAT

Two closed convex objects are not intersecting if there exists a line onto which the two objects' projections do not overlap

But which lines do I check, there are infinitely many?



SAT

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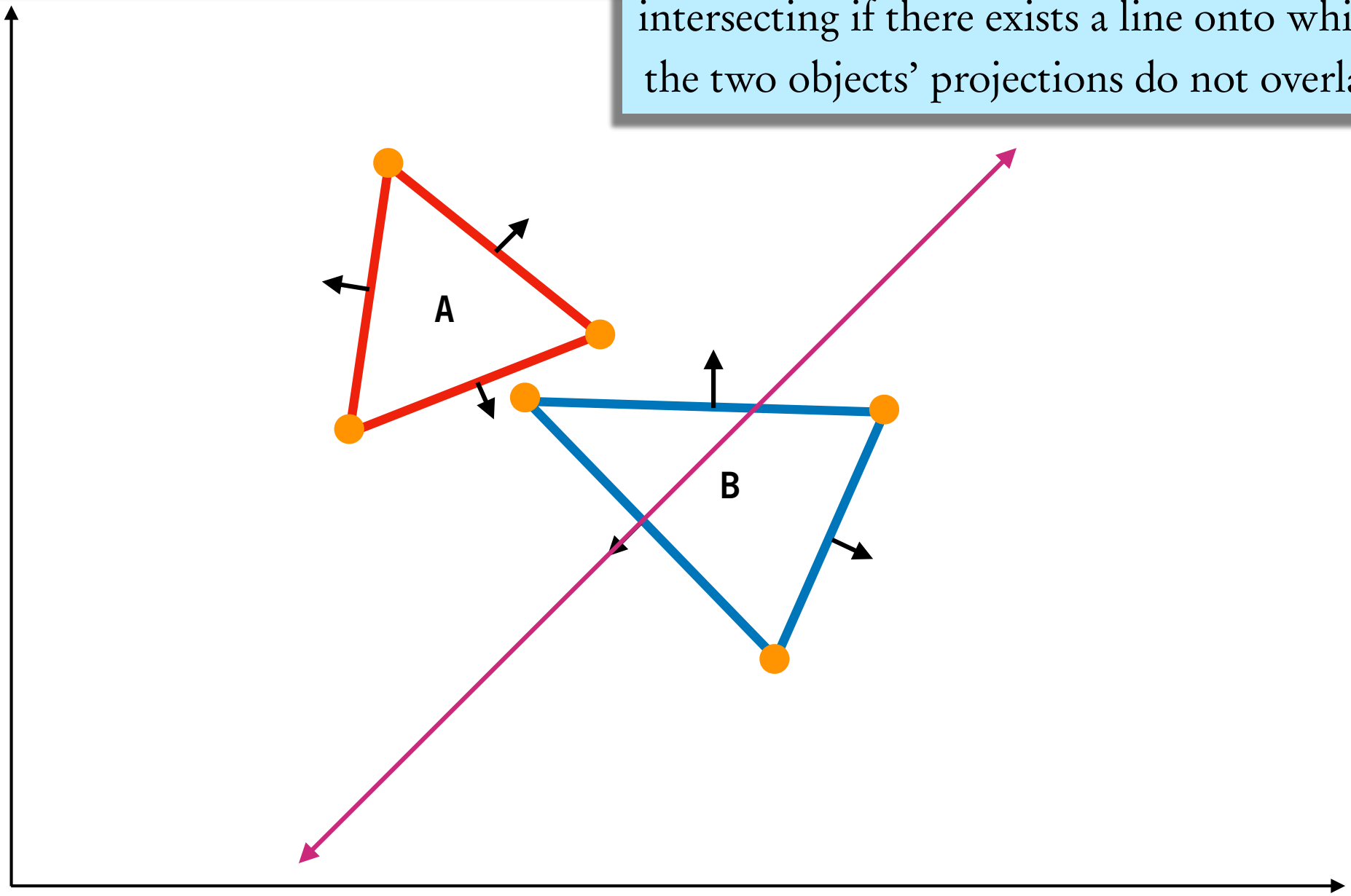

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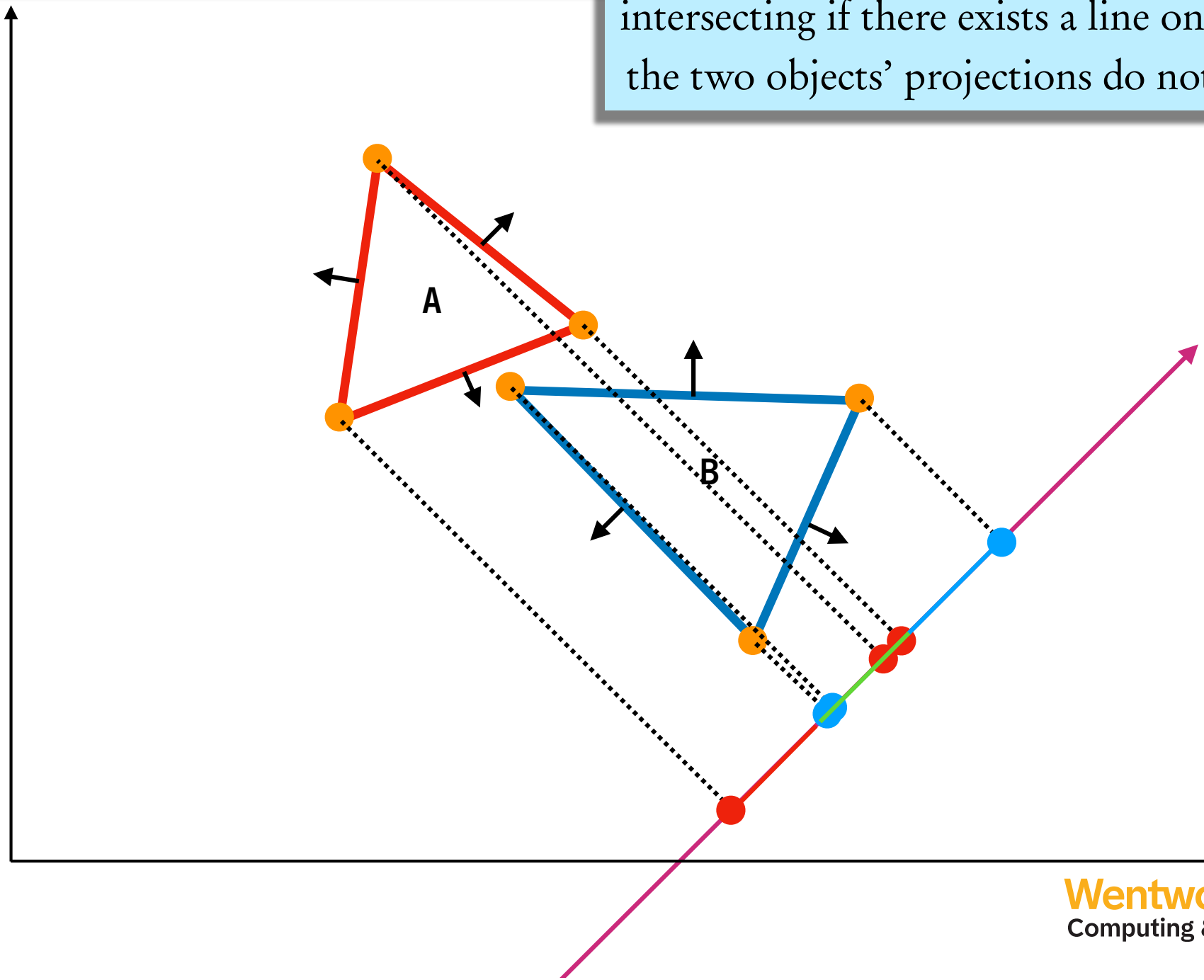

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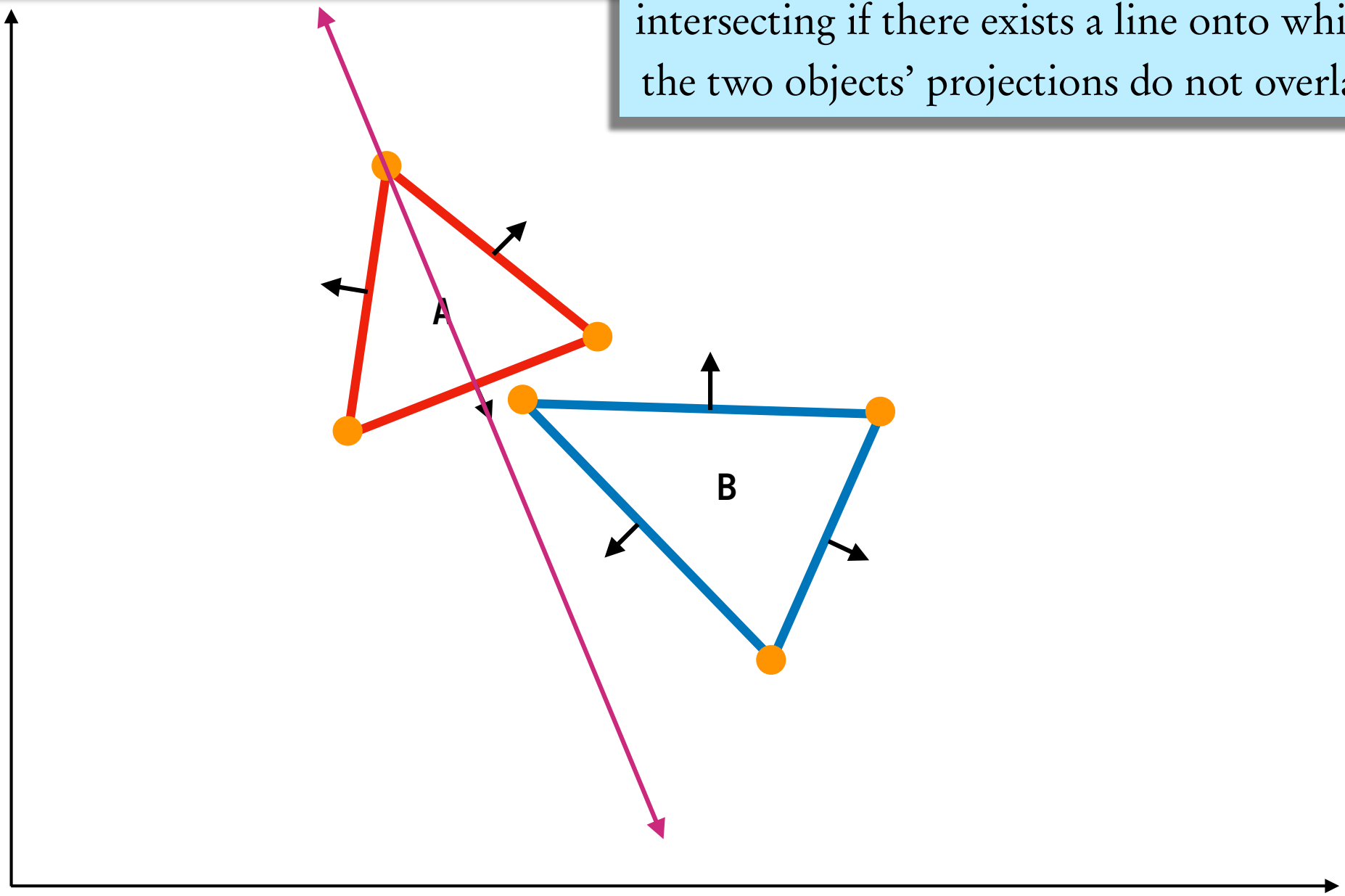

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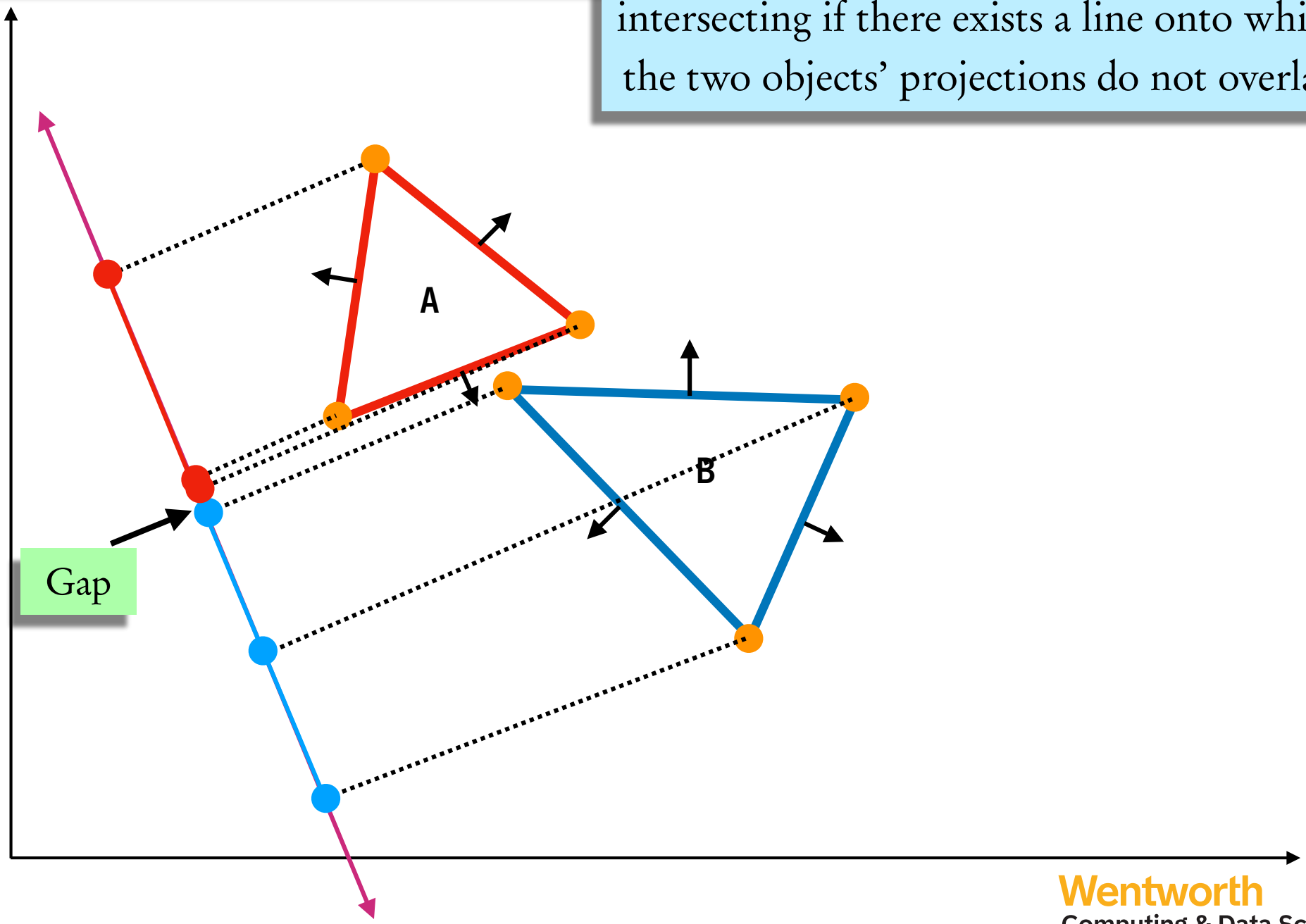
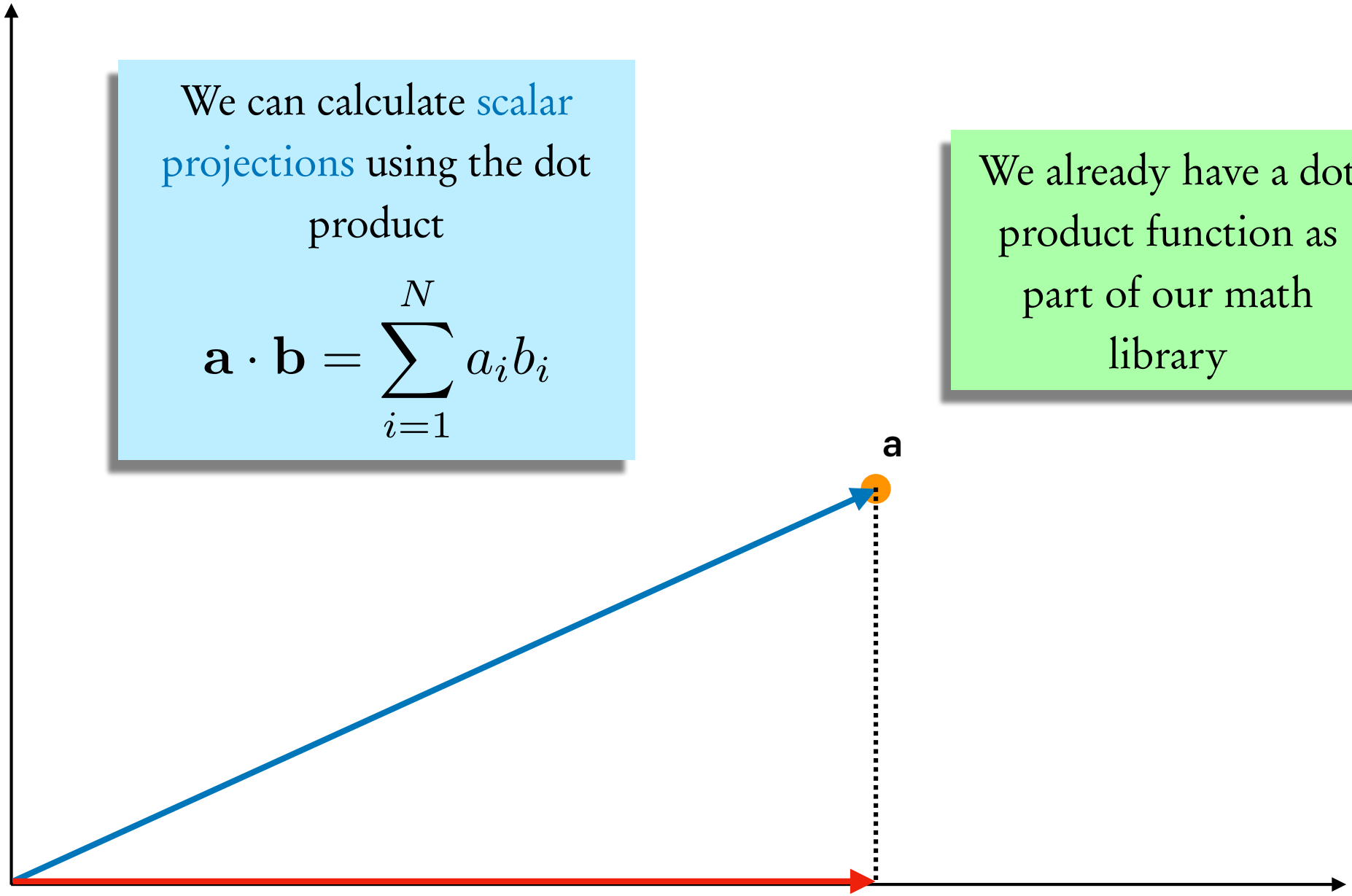


How To: Projections

We can calculate scalar projections using the dot product

$$\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^{N} a_i b_i$$

We already have a dot product function as part of our math library



SAT: Implementation

- There are several ways to implement SAT.
- I like to show the version that follows what we have just seen:

```
• Loop over all vertices of polygon A, computing the normal for
  each
• Project all vertices of A unto the normal and keep track of the
  max and min extent.
• Project all vertices of B unto the normal and keep track of the
  max and min extent.
• If the max extent of A is less than the min extent of B
                                OR
• If the max extent of B is less than the min extent of A
                 There is a separation, return
```

Next Time

- Getting more collision information from SAT
- Integrate SAT into our physics engine
- Future: GJK